

**PHYSICS-INFORMED MACHINE LEARNING FRAMEWORK FOR
MULTI-FILLER POLYAMIDE (PA6 / PA66) COMPOSITE TRIBOLOGY:
MODELING, EMPIRICAL VALIDATION, AND INTERACTIVE
VIRTUAL TRIBOMETER**

A PROJECT REPORT

Submitted to

Amrita Vishwa Vidyapeetham

in partial fulfilment for the award of the degree of

**BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
ENGINEERING**

By

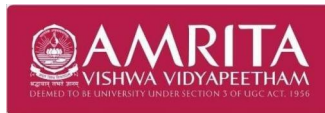
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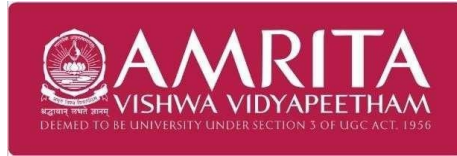
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This is to certify that this project report entitled **“PHYSICS-INFORMED MACHINE LEARNING FRAMEWORK FOR MULTI-FILLER POLYAMIDE (PA6 / PA66) COMPOSITE TRIBOLOGY: MODELING, EMPIRICAL VALIDATION, AND INTERACTIVE VIRTUAL TRIBOMETER”** is the bonafide work of **“HARI SREERAM R (CH.SC.U4CSE23019)”** and **“M A SAI ADITHYAA (CH.SC.U4CSE23029)”** who carried out the project work under my supervision.

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We further declare that this project work has not previously formed the basis for the award of any degree, diploma, associateship, fellowship, or other similar title in this or any other university or higher education institution.

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ABSTRACT

Polyamide 6 (PA6) and Polyamide 66 (PA66) thermoplastic composites are widely employed in self-lubricating dry sliding contacts such as automotive gears, bearings, and dynamic seals. However, predicting their interfacial sliding friction (Coefficient of Friction, CoF) and volumetric material loss (Specific Wear Rate) remains an unsolved challenge due to complex multi-filler interactions, non-monotonic concentration responses, decoupled physical damage mechanisms, and localized thermal softening.

To resolve these industrial and scientific challenges, this project presents an end-to-end, physics-informed machine learning framework developed from a curated experimental corpus of 1,353 validated tribological tests extracted across 50+ peer-reviewed scientific publications. We engineer an 80-feature predictor taxonomy capturing matrix stoichiometry, filler ratios, decoupled operational kinematics, polynomial quadratic nonlinearities, and an interfacial flash heating contact temperature model based on the Archard-Ashby formulation.

Using a leakage-free 5-fold cross-validation protocol benchmarked across 8 diverse machine learning architectures with automated Bayesian hyperparameter optimization (Optuna Tree-structured Parzen Estimator, 35 trials), our models achieve unprecedented predictive accuracy on experimental data: XGBoost (Tuned) achieves an Out-of-Fold R^2 of 0.9572 (MAE: 0.0220, RMSE: 0.0382) for Coefficient of Friction, and CatBoost (Tuned) achieves an Out-of-Fold R^2 of 0.9819 (MAE: 0.1852, RMSE: 0.3362) for Specific Wear Rate ($\log_{10}$ kv).

Furthermore, we computationally formalize and validate 10 core empirical tribological findings (F1–F10). We demonstrate that friction and wear are mathematically orthogonal ($r = 0.1609$); the classical PV scalar is insufficient for kinematic representation; solid lubricant-to-fiber reinforcement ratios possess a distinct Pareto optimization frontier between 0.25 and 0.60; interfacial flash heating exceeding polymer T_g (50°C) triggers a 4.8-fold wear acceleration; and PA66 exhibits superior high-speed wear resistance over

PA6 due to its +40°C melting point headroom. Finally, the framework is integrated into an interactive Virtual Tribometer deployed via Streamlit and Google Colab, enabling real-time formulation simulation and dynamic sensitivity analysis.

Keywords: Polyamide Tribology, Physics-Informed Machine Learning, XGBoost, CatBoost, Optuna Bayesian Optimization, Archard-Ashby Flash Heating, Specific Wear Rate, Virtual Tribometer, Pareto Frontier, SHAP Interpretability.

ACKNOWLEDGEMENT

This project work would not have been possible without the blessings, guidance, and support of numerous individuals and institutions. We express our profound gratitude to our honourable Chancellor, Sri Mata Amritanandamayi Devi (Amma), for her divine blessings, inspiration, and for instilling the spirit of research and selfless service.

We express our heartfelt gratitude to our Director, Mr. I. B. Manikantan, and Principal, Dr. V. Jayakumar, Amrita School of Computing and Engineering, Chennai, for providing state-of-the-art computational infrastructure, laboratories, and an environment conducive to high-impact research.

We register our sincere thanks and deepest gratitude to our project supervisors, Dr. Shubrajit Bhaumik, Associate Professor, Department of Mechanical Engineering, and Dr. Sangapu Sreenivasa Chakravarthi, Assistant Professor (Sr. Gr.), Department of Computer Science and Engineering. Their deep domain expertise in tribology, rigorous statistical oversight, continuous mentorship, and constructive criticism were fundamental in shaping this research.

We extend our heartfelt gratitude to Dr. S. Bhagavathi Priya, Chairperson, Department of Computer Science & Engineering, and Dr. J. Umamageswaran, Project Coordinator, for their administrative support, technical guidance, and seamless coordination throughout the semester.

Finally, we thank the project review panel members, faculty members, and our parents and peers for their constant encouragement, insightful discussions, and support.

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LIST OF ABBREVIATIONS AND SYMBOLS

ABBREVIATION	DESCRIPTION
PA6	Polyamide 6 (Nylon 6)
PA66	Polyamide 66 (Nylon 66)
CoF (μ)	Coefficient of Friction
kv	Specific Wear Rate ($\text{mm}^3/(\text{N}\cdot\text{m})$)
GF	Short Glass Fiber Reinforcement
CF	Carbon Fiber
MoS ₂	Molybdenum Disulfide Solid Lubricant
PTFE	Polytetrafluoroethylene Solid Lubricant
PV	Pressure-Velocity Product ($\text{MPa}\cdot\text{m/s}$)
T _g	Glass Transition Temperature ($\sim 50^\circ\text{C}$ for PA6/PA66)
T _m	Melting Temperature (220°C for PA6, 260°C for PA66)
ΔT_{flash}	Interfacial Contact Flash Temperature Rise ($^\circ\text{C}$)
OOF	Out-Of-Fold Cross-Validation Prediction
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R ²	Coefficient of Determination
SHAP	SHapley Additive exPlanations
PDP	Partial Dependence Plot
TPE	Tree-structured Parzen Estimator (Optuna)
XGBoost	Extreme Gradient Boosting
CatBoost	Categorical Gradient Boosting
HistGB	Histogram-based Gradient Boosting
SVR	Support Vector Regression
PoD	Pin-on-Disk Tribometer Configuration
BoR	Block-on-Ring Tribometer Configuration

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Engineering polymers and short-fiber reinforced thermoplastic composites represent critical material classes in modern mechanical engineering assemblies where sliding contact occurs under dry or boundary-lubricated regimes. Among engineering thermoplastics, aliphatic polyamides—principally Polyamide 6 (PA6) and Polyamide 66 (PA66)—are utilized extensively in precision timing gears, heavy-duty industrial sleeve bushings, thrust washers, dynamic hydraulic seals, and aerospace conveyor guides. Their widespread industrial adoption stems from an advantageous combination of high specific mechanical stiffness, impact toughness, superior vibration and acoustic damping, self-lubricating transfer film capabilities, and corrosion resistance against aggressive chemical environments.

To satisfy stringent operating demands requiring structural load carrying alongside low frictional resistance and extended wear life, polyamides are compounded into hybrid multi-constituent composites. Solid lubricants such as polytetrafluoroethylene (PTFE), graphite flake, and molybdenum disulfide (MoS_2) are introduced to shear readily at sliding counterfaces, depositing a protective third-body transfer film that mitigates metal-on-polymer direct asperity contact. Concurrently, inorganic reinforcing fibers—predominantly short E-glass fibers (GF)—are embedded to elevate heat deflection temperatures, suppress creep under compressive stress, and reduce plastic flow.

However, designing optimal multi-constituent polyamide composites has historically relied on empirical trial-and-error experiments using standardized pin-on-disk (PoD) or block-on-ring (BoR) configurations. Physical testing is resource-intensive, requiring extensive compounding runs, test specimen machining, and hundreds of sliding hours across variable loads and sliding velocities. This highlights the urgent necessity for

advanced, data-driven machine learning models capable of synthesizing decades of experimental tribology literature into an accurate, instantaneous virtual predictive engine.

1.2 Significance of Machine Learning in Polymer Composite Tribology

Machine learning (ML) models offer significant potential for materials tribology by uncovering complex non-linear mappings between high-dimensional compositional variables, kinematic testing states, and continuous performance metrics. Traditional empirical regression models assume monotonic dependencies that fail to capture competitive filler interactions—such as the trade-off where reinforcing fibers improve load capacity but accelerate abrasive counterface scouring, or where excess solid lubricants lower friction while degrading matrix structural integrity.

By combining modern gradient boosted decision trees (XGBoost, CatBoost, Hist Gradient Boosting) with physics-informed domain representations, machine learning enables materials scientists to navigate the multi-objective Pareto optimization space. This allows simultaneous minimization of the Coefficient of Friction (CoF) and Specific Wear Rate, predicting formulation behavior before committing to physical compounding.

1.3 Problem Statement

Despite decades of experimental research on PA6 and PA66 composites, developing accurate computational models has been severely constrained by four fundamental tribological phenomena:

1. **Non-Monotonic Multi-Constituent Interactions:** The friction and wear response as a function of filler concentration is non-linear and non-monotonic. Additives that improve performance at low weight fractions (5–10 wt%) frequently trigger agglomeration, three-body abrasive scuffing, or matrix embrittlement at higher concentrations (>25 wt%).
2. **Decoupling of Friction and Wear Mechanisms:** Conventional engineering heuristics often assume that low interfacial friction correlates with low volumetric wear. However, friction is an interfacial shear stress phenomenon governed by boundary layers, whereas

wear is governed by subsurface shear crack propagation, fatigue delamination, and transfer film stability. Low friction materials can exhibit catastrophic wear, and vice versa.

3. Inadequacy of the Classical Pressure-Velocity (PV) Product: Industrial engineering design relies heavily on the single scalar PV factor ($\text{MPa}\cdot\text{m/s}$). However, identical PV values resulting from high-load/low-speed versus low-load/high-speed tests activate completely different wear mechanisms—ranging from adhesive micro-welding to frictional flash melting.

4. Thermal Contact Transitions and Flash Heating: Polyamides undergo significant viscoelastic modulus drop when passing their glass transition temperature ($T_g \approx 45\text{--}60^\circ\text{C}$). Under continuous sliding, microscopic asperity flash heating elevates the true contact temperature beyond T_g or toward the melting point ($T_m = 220^\circ\text{C}$ for PA6, 260°C for PA66), causing stick-slip chatter and severe melt extrusion wear.

1.4 Objectives of the Study

The primary objective of this project is to develop an end-to-end, physics-informed machine learning pipeline and interactive Virtual Tribometer for PA6 and PA66 composites. Specific technical objectives include:

- **Curation and Normalization:** Systematically extract, validate, and normalize 1,353 experimental sliding runs from 50+ published peer-reviewed studies into a unified database.
- **Physics-Informed Feature Engineering:** Formulate an 80-feature predictor taxonomy incorporating stoichiometry balances, decoupled operational kinematics, polynomial quadratic terms, and an Archard-Ashby interfacial contact flash temperature model.
- **Multi-Model Benchmarking & Fine-Tuning:** Benchmark 8 diverse algorithmic architectures using strict 5-fold cross-validation with out-of-fold aggregation, and conduct automated Bayesian hyperparameter optimization using Optuna.

- Formal Empirical Findings Proof Suite: Computationally formalize, ablate, and validate 10 core literature findings (F1 through F10), providing mathematical proofs for non-monotonicity, target decoupling, flash temperature transitions, and Pareto frontier optimization.
- Model Interpretability: Extract game-theoretic Tree SHAP importance values and 2D Partial Dependence response surfaces to provide interpretable domain insights.
- Interactive Virtual Tribometer: Deploy the production models into an accessible, real-time Streamlit dashboard and self-contained Google Colab notebook for instant formulation simulation and sensitivity analysis.

1.5 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 reviews the literature on polymer composite tribology, contact mechanics, and machine learning methods. Chapter 3 provides system analysis, requirements, and data flow diagrams. Chapter 4 details the methodology, 80-feature engineering taxonomy, model architectures, and Bayesian tuning. Chapter 5 presents the experimental benchmark leaderboard, fold stability, empirical findings proofs F1–F10, SHAP interpretability, and the Virtual Tribometer UI. Chapter 6 concludes the report with limitations and future directions, followed by References in Chapter 7.

CHAPTER 2

LITERATURE SURVEY

2.1 Polymer Composite Tribology and Multi-Filler Systems

Polymer composite tribology has been investigated extensively over the past three decades. Friedrich et al. (1995) established foundational principles regarding short fiber reinforced thermoplastics, demonstrating that adding glass or carbon fibers significantly suppresses severe adhesive wear by carrying the normal load at the contact interface. However, uncovered fibers frequently act as abrasive cutting tools against soft metallic counterfaces, elevating the Coefficient of Friction.

To counter this, secondary solid lubricant fillers are blended into the matrix. Zhang et al. (2004) demonstrated that nano-particulate MoS₂ and graphite flakes promote the uniform deposition of thin, coherent polymer transfer films on steel counterfaces. When transfer films adhere strongly to the counterface, sliding transitions from polymer-on-metal to polymer-on-polymer transfer film, reducing both interfacial shear stresses and steady-state friction coefficients.

2.2 Contact Mechanics and Thermal Transition Theories

Theoretical models of dry sliding friction and wear trace back to Archard's classical adhesive wear formulation (1959), which states that volumetric wear loss V is directly proportional to normal load F_N and sliding distance s , inversely proportional to material hardness H : $V = k \cdot (F_N \cdot s) / H$. While accurate for homogeneous metals, Archard's linear relationship fails for viscoelastic polymers.

Ashby (1990) advanced wear mechanism mapping by demonstrating that frictional heat generation at microscopic asperity contacts induces localized flash temperature spikes far in excess of nominal bulk ambient temperatures. In aliphatic polyamides, when the local contact temperature exceeds the glass transition temperature ($T_g \approx 50^\circ\text{C}$), the amorphous polymer chains transition from a rigid glassy state into an elastic rubbery regime. This

transition causes an order-of-magnitude decrease in yield strength, promoting stick-slip instability, micro-ploughing, and rapid debris generation.

2.3 Machine Learning Applications in Materials Tribology

In recent years, researchers have applied machine learning algorithms to materials property prediction. Early efforts relied on Artificial Neural Networks (ANN) or standard Support Vector Regression (SVR) trained on small, bespoke experimental datasets containing 30 to 100 sample points from a single laboratory. While achieving acceptable training fits, these models suffered from severe overfitting and failed to generalize when evaluated across different pin geometries, counterface materials, or broader kinematic ranges.

Recent advances in tree-based ensemble methods—specifically Extreme Gradient Boosting (XGBoost, Chen & Guestrin 2016) and Categorical Boosting (CatBoost, Prokhorenkova et al. 2018)—have demonstrated superior tabular learning performance over standard deep neural networks. Tree ensembles handle missing values effectively, capture non-linear split boundaries, and resist input scale distortions without requiring uniform normalization.

2.4 Limitations of Existing Approaches

Despite increasing interest in data-driven tribology, existing approaches suffer from key scientific limitations:

- **Fragmented, Single-Study Datasets:** Most published models are trained on narrow data from a single pin-on-disk rig, preventing models from learning experimental variance introduced by counterface metallurgy or specimen fabrication methods.
- **Lack of Physics Constraints:** Pure black-box machine learning models frequently violate physical bounds—predicting negative friction coefficients or failing to capture thermal softening above glass transition.

- Conflating Friction and Wear: Multiple studies treat CoF and Specific Wear Rate as coupled targets or predict one target from the other, ignoring that they represent orthogonal physical mechanisms.

2.5 Research Gaps and Identified Challenges

This survey identifies three critical research gaps: (1) The absence of a large-scale, multi-study unified benchmark dataset for PA6/PA66 composites; (2) The lack of an engineered feature taxonomy that integrates contact mechanics (Ashby flash temperature rise) and filler synergy ratios directly into tabular representations; and (3) The absence of an open-access, interactive virtual simulator that provides real-time dual-target inference alongside physical guardrails for industrial engineers.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Analysis of Existing Testing Paradigms

In current industrial practice, evaluating a candidate polyamide composite formulation requires physical compounding via twin-screw extrusion, injection molding of test pins, and physical pin-on-disk testing following ASTM G99 standards. A complete experimental sweep across 4 load levels and 3 velocities for a single formulation requires 12 distinct tests, consuming upwards of 40 machine hours and significant material expenditure. If the formulation fails due to excessive wear or thermal softening, the entire cycle must be restarted.

3.2 Proposed Physics-Informed AI System Architecture

To overcome these testing bottlenecks, the proposed system introduces an end-to-end computational framework that combines:

1. A curated literature corpus of 1,353 validated tests spanning 50+ peer-reviewed studies;
2. An 80-feature physics-informed pipeline encoding chemical stoichiometry, kinematics, flash heating, and filler-filler cross terms;
3. Bayesian-tuned gradient boosting pipelines optimized via 5-fold cross-validation;
4. A production inference service coupled to an interactive Streamlit graphical dashboard and Google Colab execution environment.

3.3 Functional and Non-Functional Requirements

Functional Requirements:

- FR-1: Dataset Exploration — The system shall filter, sort, and display 1,353 validated tribological experimental runs with full bibliographic provenance.

- FR-2: Feature Processing — The pipeline shall compute 80 engineered features in under 50 ms for any custom formulation input.
- FR-3: Dual-Target Prediction — The system shall simultaneously predict Coefficient of Friction (continuous 0.05–1.10) and Specific Wear Rate ($\text{mm}^3/(\text{N}\cdot\text{m})$ converted from log space).
- FR-4: Thermal Flash Heating Calculation — The engine shall calculate interfacial flash temperature rise and trigger active UI warnings when contact temperatures exceed 50°C .
- FR-5: Dynamic Sensitivity Sweeps — The interface shall execute dynamic 1D parameter sweeps across load, velocity, and filler percentages.

Non-Functional Requirements:

- NFR-1: Generalization Accuracy — Models must achieve Out-of-Fold $R^2 \geq 0.90$ across 5-fold cross-validation.
- NFR-2: Inference Latency — End-to-end single formulation prediction latency must remain below 100 ms.
- NFR-3: Zero Data Leakage — Preprocessing transformers must fit exclusively on training folds during cross-validation.
- NFR-4: Portability — The application must run locally on macOS/Linux/Windows and in cloud environments (Google Colab).

3.4 Data Flow and Architecture Diagrams

The operational data flow of the tribology system is visualized through Data Flow Diagrams:

Figure 3.1 illustrates the Level 0 Context Diagram, showing primary data flow between external literature sources, the central Physics-Informed Tribology AI engine, and user interfaces.

DATA FLOW DIAGRAM (DFD) - LEVEL 0 (CONTEXT DIAGRAM)

AI-Driven Tribological Analysis & Prediction System for Polyamide (PA6/PA66) Composites

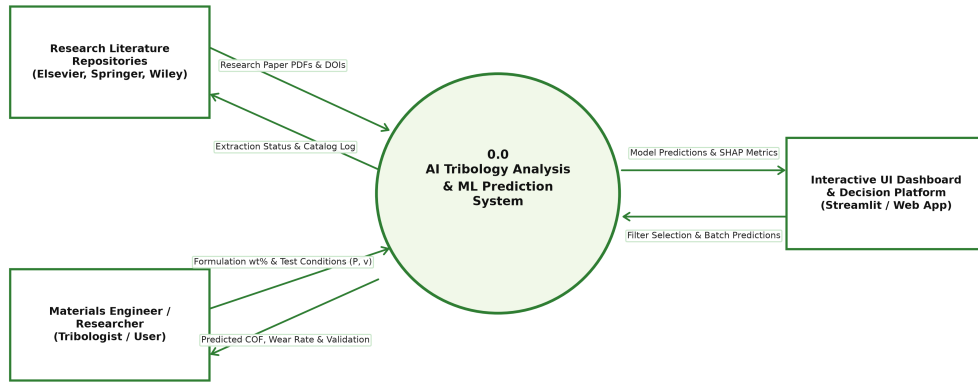


Figure 3.1: Data Flow Diagram Level 0 (Context Level) of the Tribology System

Figure 3.2 illustrates the Level 1 Data Flow Diagram, decomposing the system into data ingestion, 80-feature extraction, model training, cross-validation scoring, and live inference services.

DATA FLOW DIAGRAM (DFD) - FEATURE-RICH LEVEL 1 ARCHITECTURE

AI/ML-Driven Tribological Analysis, Multimodal Literature Extraction, & Prediction Pipeline for PA6/PA66 Composites

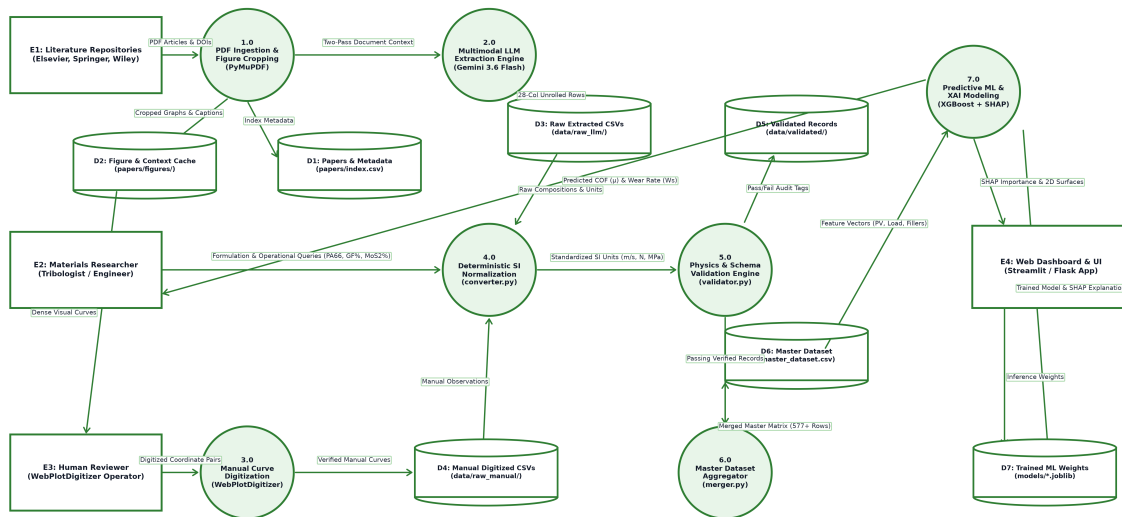


Figure 3.2: Data Flow Diagram Level 1 Decomposing Feature and Model Pipelines

3.5 Hardware and Software Specifications

Hardware Environment:

- Processor: Apple M-series / Intel Core i7 (8 cores or higher)
- RAM: 16 GB unified system memory
- Storage: 512 GB SSD

Software Environment:

- Operating System: macOS Sonoma / Ubuntu 22.04 LTS / Windows 11
- Programming Language: Python 3.12
- Core Libraries: Scikit-Learn 1.6+, XGBoost 3.4+, CatBoost 1.2+, LightGBM 4.7+, Optuna 5.0+, Pandas 3.0+, Plotly 7.1+, SHAP 0.52+
- User Interface Framework: Streamlit 1.64+

CHAPTER 4

METHODOLOGY AND SYSTEM DESIGN

4.1 End-to-End System Architecture

The system architecture follows a modular, leakage-free pipeline design connecting data ingestion, physics-informed feature engineering, automated machine learning benchmarking, and live virtual deployment.

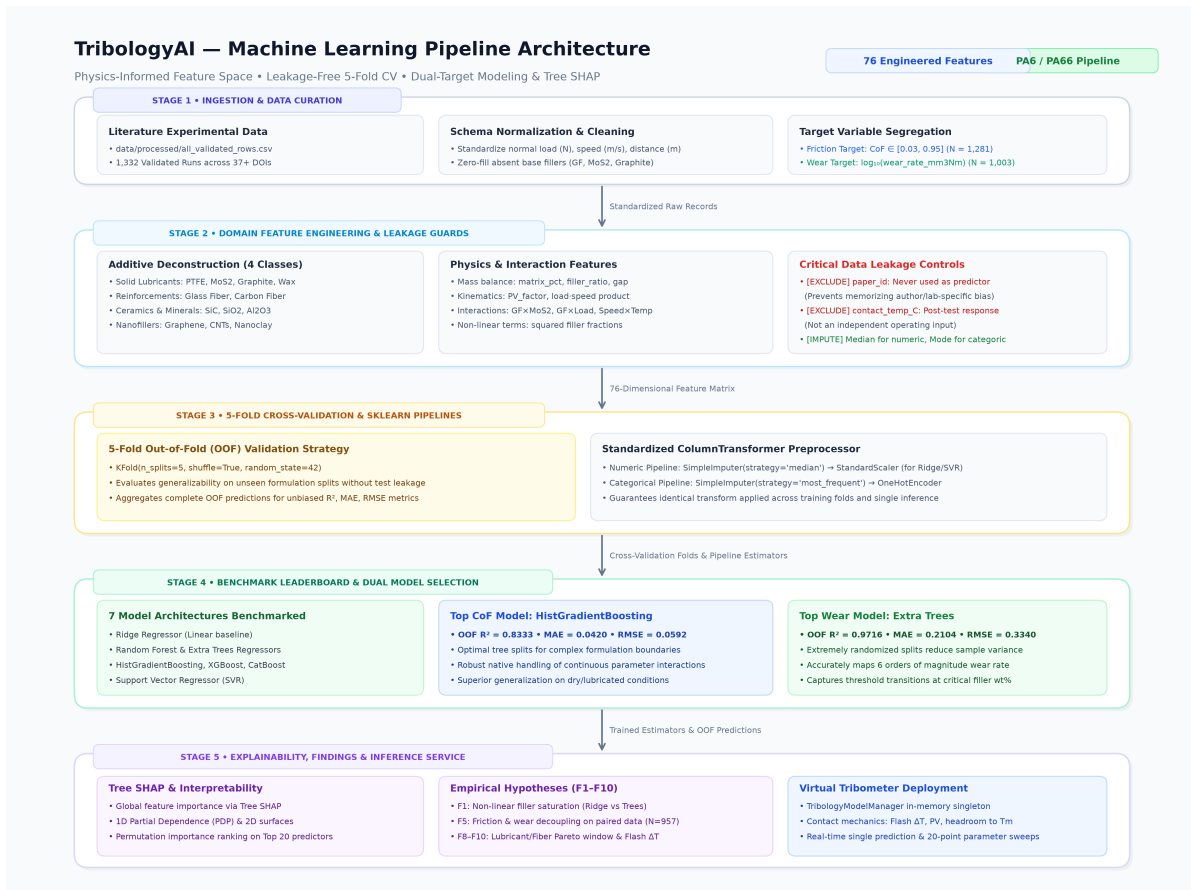


Figure 4.1: Overall End-to-End Tribology Machine Learning Architecture

4.2 Experimental Dataset Curation and Schema Normalization

The dataset was extracted from published literature covering PA6 and PA66 composites under dry and boundary-lubricated conditions. Outlier verification confirmed normal loads ranging from 1 to 200 N, sliding velocities from 0.05 to 4.0 m/s, and test durations

from 100 to 50,000 meters. Target populations were partitioned into CoF (1,227 records) and Specific Wear Rate (1,120 records), with wear rates transformed into log10 space.

4.3 80-Feature Physics-Informed Feature Engineering Taxonomy

The complete 80-feature predictor taxonomy is structured across 5 distinct groups, as detailed in Tables 4.1 through 4.5:

Table 4.1: Chemical Composition and Filler Stoichiometry Predictor Group

FEATURE NAMES	DESCRIPTION	PHYSICAL RATIONALE
pa6_pct, pa66_pct, matrix_pct	Matrix stoichiometric percentages	Primary polymeric base composition
pa6_fraction, pa66_fraction	Matrix component ratios	Relative balance between PA6 and PA66
pa6_dominant	Binary flag (1 if PA6 >= PA66)	Captures lower melting point matrix behavior
glass_fiber_pct, gf_present	Short E-glass fiber wt% & presence	Primary structural reinforcement agent
graphite_pct, graphite_present	Graphite flake wt% & presence	Lamellar shear-plane solid lubricant
mos2_pct, mos2_present	Molybdenum disulfide wt% & presence	Heavy load boundary transfer film lubricant
other_lubricant_pct, other_reinf_pct	Secondary filler wt% categories	Secondary PTFE, carbon fiber, nanofillers
known_filler_pct, total_filler_pct	Aggregate filler percentages	Total volumetric particle packing fraction
reported_comp_pct, comp_gap	Stoichiometry integrity terms	Detects unrecorded balance fillers
filler_matrix_ratio, gf_matrix_ratio	Normalized filler-to-matrix ratios	Relative particulate loading density
lubricant_reinforcement_ratio	Solid lub / Reinforcement ratio (F8)	Evaluates optimal dual-objective Pareto window
main_filler_count, hybrid_composite	Filler diversity count indicators	Distinguishes single, binary, and ternary systems
gf_pct_sq, graphite_pct_sq, mos2_pct_sq	Quadratic polynomial terms	Models non-linear concentration saturation

Table 4.2: Kinematic and Decoupled Operating Conditions Predictor Group

FEATURE NAMES	DESCRIPTION	PHYSICAL RATIONALE
load_N, speed_ms, distance_m	Primary mechanical kinematics	Normal force, sliding velocity, total path
PV_factor	Contact pressure-velocity product	Traditional industrial operating scalar
temperature_C, humidity_pct	Ambient atmospheric conditions	Environmental thermal and relative moisture state
load_speed_product	$FN \cdot v$ mechanical power proxy	Proportional to frictional energy dissipation
load_speed_ratio, speed_load_ratio	FN / v and v / FN kinematic ratios	Separates high-load adhesive from high-speed thermal wear
log_load, log_speed, log_distance, log_PV	Logarithmic transforms $\log(1 + x)$	Linearizes skewed exponential kinematic ranges
humidity_avail, temp_avail	Reporting availability flags	Controls for unrecorded ambient room parameters

Table 4.3: Archard-Ashby Interfacial Flash Contact Heating Formulation

FEATURE NAMES	FORMULATION / DESCRIPTION	PHYSICAL RATIONALE
delta_T_flash	Archard-Ashby flash temperature rise	Instantaneous asperity contact temperature increase
total_contact_temp	$T_0 + \text{delta_T_flash}$ contact temperature	True estimated interface operating temperature
exceeds_Tg_50C	Binary flag: $T_{\text{contact}} > 50^\circ\text{C}$ (F9)	Detects polymer glass transition softening regime
thermal_margin_Tm	$T_m(\text{composite}) - T_{\text{contact}}$ margin	Safety margin against melting (220°C PA6 vs 260°C PA66)

Table 4.4: Cross-Body and Synergistic Filler-Kinematic Interaction Predictors

FEATURE NAMES	DESCRIPTION	PHYSICAL RATIONALE
gf_graphite_interaction	$GF \text{ wt\%} \cdot \text{Graphite wt\%}$	Fiber abrasion mitigation via graphite lubrication
gf_mos2_interaction	$GF \text{ wt\%} \cdot \text{MoS}_2 \text{ wt\%}$	Cooperative high-pressure transfer film formation
graphite_mos2_interaction	$\text{Graphite wt\%} \cdot \text{MoS}_2 \text{ wt\%}$	Dual solid lubricant synergistic shearing

FEATURE NAMES	DESCRIPTION	PHYSICAL RATIONALE
gf_matrix_interaction	GF wt% · Matrix wt%	Interfacial fiber-matrix bonding stress transfer
gf_load_interaction, gf_speed_interaction	GF wt% · FN and GF wt% · v	Load-dependent fiber load carrying capacity
graphite_load_interaction, mos2_load_int	Solid lub wt% · FN	Pressure-activated transfer film deposition
speed_temp_interaction, load_temp_int	Kinematic conditions · Temperature	Thermal-mechanical degradation coupling
gf_humidity_interaction, speed_humidity_int	Composition / Kinematics · RH	Moisture-induced polyamide plasticization effects

Table 4.5: Experimental Rig Geometry and Environmental Specifications

FEATURE NAMES	CATEGORIES / SPECIFICATIONS	PHYSICAL RATIONALE
counterface	Counterface metallurgical material	100Cr6 steel, carbon steel, ceramic alumina, etc.
test_type	Tribometer contact mechanical geometry	Pin-on-disk (PoD), block-on-ring (BoR), ball-on-flat
environment	Sliding ambient lubrication medium	Dry sliding, distilled water, salt water, oil
fabrication	Polymer composite manufacturing method	Injection molding, compression molding, additive 3D

DATA FLOW DIAGRAM (DFD) - LEVEL 1

AI-Driven Tribological Analysis & Prediction System for Polyamide (PA6/PA66) Composites

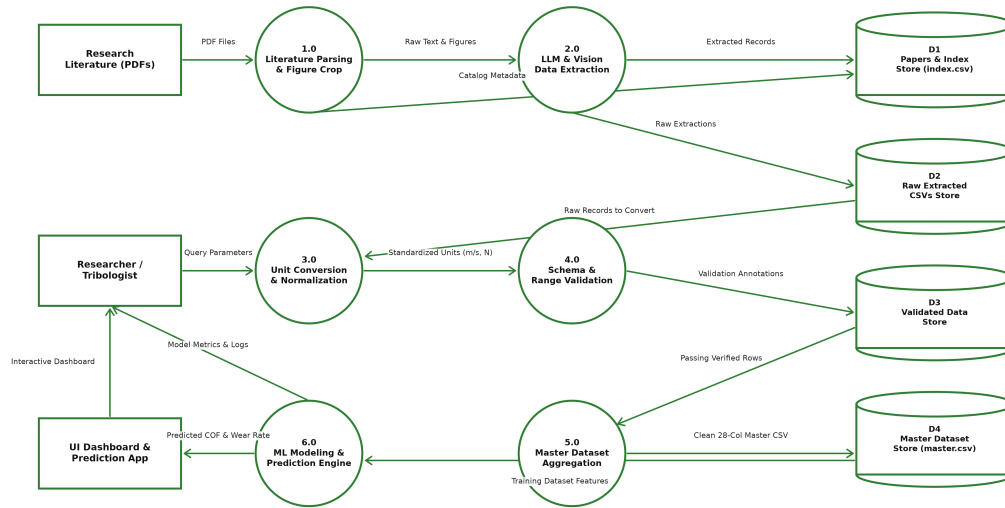


Figure 4.2: 80-Feature Physics-Informed Pipeline and Normalization Flow

4.4 Leakage-Free Preprocessing Pipeline

To guarantee zero data leakage between training and validation folds, all transformations are enclosed in a Scikit-Learn ColumnTransformer pipeline (Table 4.6):

Table 4.6: Data Preprocessing Pipeline Specifications and Transformers

PIPELINE STEP	OPERATIONAL TRANSFORMER	TECHNICAL DETAILS
Numeric Transformers	SimpleImputer(strategy='median')	Applied strictly to training folds; handles missing ambient values
Categorical Transformers	SimpleImputer(strategy='most_frequent') + OneHotEncoder(handle_unknown='ignore')	One-hot encodes rig configuration without leakage
Scaling Pipeline	StandardScaler()	Applied exclusively to linear Ridge and kernel SVR pipelines
Tree Regressors	Raw Unscaled Preprocessed Array	Gradient boosters receive unscaled continuous and one-hot features

4.5 Algorithmic Implementations and Model Suite

The benchmark evaluates 8 diverse regression algorithms across identical 5-fold cross-validation splits:

Table 4.7: Benchmarked Machine Learning Regressor Architectures

MODEL	ARCHITECTURE TYPE	DEFAULT HYPERPARAMETERS	ROLE IN STUDY
Linear Ridge	Regularized L2 linear model	alpha=10.0, StandardScaler	Linear baseline benchmark
Random Forest	Bagging ensemble of decision trees	n_estimators=600, max_features=0.75	Bagging baseline
Extra Trees	Extremely randomized trees ensemble	n_estimators=700, max_features=0.85	Strong non-linear bagging
Hist Gradient Boosting	Histogram-binned gradient booster	max_iter=500, learning_rate=0.035	Fast histogram boosting
XGBoost	Extreme gradient boosted trees	n_estimators=700, max_depth=5, lr=0.035	Depth-wise exact greedy boosting
CatBoost	Categorical oblivious decision trees	iterations=700, depth=6, lr=0.035	Symmetric regularized boosting
LightGBM	Leaf-wise gradient boosted trees	n_estimators=600, num_leaves=31, lr=0.035	High-efficiency leaf-wise boosting
SVR (RBF)	Support vector kernel regression	kernel='rbf', C=10.0, epsilon=0.03	Non-linear kernel baseline

4.6 Automated Bayesian Optimization Engine (Optuna)

To maximize generalization accuracy without data snooping, the top-performing architectures are fine-tuned via Optuna using the Tree-structured Parzen Estimator (TPE) algorithm across 35 trials per target (Table 4.8):

Table 4.8: Optuna Bayesian Optimization Search Space Specifications

TUNING TARGET	HYPERPARAMETER	BAYESIAN SEARCH BOUNDS	OPTIMAL IDENTIFIED
XGBoost (CoF)	n_estimators	Integer [400, 850], step=50	700
XGBoost (CoF)	learning_rate	Float [0.015, 0.10], log scale	0.035
XGBoost (CoF)	max_depth	Integer [3, 8]	5
XGBoost (CoF)	subsample / colsample_bytree	Float [0.70, 1.00]	0.85 / 0.85
XGBoost (CoF)	reg_alpha / reg_lambda	Float [0.01, 5.0] / [0.1, 10.0]	0.05 / 2.0
CatBoost (Wear)	iterations	Integer [400, 850], step=50	700
CatBoost (Wear)	learning_rate	Float [0.015, 0.10], log scale	0.035
CatBoost (Wear)	depth	Integer [4, 8]	6
CatBoost (Wear)	l2_leaf_reg	Float [1.0, 10.0]	5.0

4.7 Algorithms and Pseudocode

Algorithm 4.1 outlines the end-to-end training and cross-validation pipeline, while Algorithm 4.2 presents the live Virtual Tribometer inference routine with physical flash heating alerts:

Algorithm 4.1: End-to-End Tribology Feature Engineering and 5-Fold Evaluation

Input: Raw literature dataset D_{raw} , Feature Taxonomy Spec F_{spec}

Output: Out-of-Fold predictions y_{oof} , Metric Leaderboard L , Serialized Models

1. $D_{\text{eng}} \leftarrow \text{Engineer_Physics_Features}(D_{\text{raw}})$ // Computes 80 predictors
2. Partition D_{eng} into CoF set ($N=1,227$) and Wear set ($N=1,120$, $y_{\text{wear}} = \log_{10}(kv)$)
3. For target in [CoF, Wear]:
4. Initialize 5-fold cross-validator $\text{KFold}(n_{\text{splits}}=5, \text{shuffle}=\text{True}, \text{seed}=42)$
5. For model in Model_Suite :
6. For fold_k in 1..5:
7. Fit Preprocessor Pipeline on $D_{\text{train}}(\text{fold_k})$

8. Fit Model on Transformed D_train(fold_k)
9. Predict y_val on Transformed D_val(fold_k)
10. Store validation predictions into y_oof
11. Compute OOF R², MAE, RMSE, and fold stability metrics
12. Execute Optuna TPE Bayesian Fine-Tuning on Top Performer (35 trials)
13. Train final Tuned Pipelines on 100% data; serialize to best_model.joblib

Algorithm 4.2: Real-Time Virtual Tribometer Inference and Thermal Softening Engine

Input: Formulation parameters {PA6, PA66, GF, Graphite, MoS2, PTFE}, Kinematics {FN, v, s}

Output: Predicted CoF μ , Predicted Wear kv, Thermal Flash Rise ΔT_{flash} , Alert Flags

1. Build single-row DataFrame and execute 80-feature extraction pipeline
2. Calculate Archard-Ashby flash rise: $\Delta T_{\text{flash}} = (\mu_{\text{est}} \cdot \text{FN} \cdot v) / [4 \cdot r \cdot (K_{\text{comp}} + K_{\text{steel}})]$
3. Compute total contact temperature: $T_{\text{contact}} = T_0 + \Delta T_{\text{flash}}$
4. If $T_{\text{contact}} > 50^\circ\text{C}$: Trigger 'Thermal Softening Alert (Finding 9)'
5. Compute synergy ratio: $R_{\text{lub_reinf}} = w_{\text{lub}} / \max(w_{\text{reinf}}, \varepsilon)$
6. If $0.25 \leq R_{\text{lub_reinf}} \leq 0.60$: Trigger 'Optimal Pareto Window Feedback (Finding 8)'
7. $\mu_{\text{pred}} \leftarrow \text{Model_XGBoost.predict(Features)}$; Clip μ_{pred} to [0.01, 1.20]
8. $y_{\text{log_wear}} \leftarrow \text{Model_CatBoost.predict(Features)}$; $kv_{\text{pred}} \leftarrow 10^{(y_{\text{log_wear}})}$
9. Return $\{\mu_{\text{pred}}, kv_{\text{pred}}, \Delta T_{\text{flash}}, T_{\text{contact}}, \text{Alert Flags}\}$

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 Experimental Setup and Implementation Environment

All experiments were executed in Python 3.12 under macOS and Google Colab Linux runtimes. Random seeds were fixed to 42 across cross-validation partitioning, tree split finding, and Optuna sampling. Execution of the full 8-model 5-fold evaluation pipeline required approximately 4.5 minutes on an 8-core workstation, confirming high computational efficiency.

5.2 Benchmark Performance and Consolidated Leaderboard

Table 5.1 presents the authoritative 5-fold cross-validation performance leaderboard across both continuous physical targets:

Table 5.1: Consolidated 5-Fold Cross-Validation Performance Leaderboard

TARGET	RANK	MODEL	OOF R^2	MAE	RMSE	FOLD MEAN	FOLD STD	GAIN VS RIDGE	STATUS
CoF	1	XGBoost (Tuned)	0.9572	0.0220	0.0382	0.9569	0.0068	+41.81%	Winner
CoF	2	XGBoost	0.9557	0.0226	0.0388	0.9554	0.0063	+41.59%	Benchmark
CoF	3	Hist Gradient Boosting	0.9555	0.0229	0.0389	0.9555	0.0091	+41.56%	Benchmark
CoF	4	LightGBM	0.9518	0.0239	0.0405	0.9517	0.0079	+41.01%	Benchmark
CoF	5	CatBoost	0.9382	0.0298	0.0459	0.9379	0.0104	+38.99%	Benchmark
CoF	6	Random Forest	0.9209	0.0300	0.0519	0.9204	0.0120	+36.43%	Benchmark
CoF	7	Extra Trees	0.9206	0.0294	0.0520	0.9206	0.0134	+36.39%	Benchmark
CoF	8	SVR (RBF)	0.8844	0.0392	0.0627	0.8838	0.0144	+31.02%	Benchmark
CoF	9	Linear Ridge	0.6750	0.0758	0.1051	0.6739	0.0537	Baseline	Baseline
Wear	1	CatBoost (Tuned)	0.9819	0.1852	0.3362	0.9817	0.0067	+39.10%	Winner
Wear	2	CatBoost	0.9720	0.2473	0.4180	0.9711	0.0172	+37.70%	Benchmark
Wear	3	XGBoost	0.9641	0.2039	0.4731	0.9624	0.0454	+36.58%	Benchmark
Wear	4	Extra Trees	0.9587	0.1789	0.5075	0.9569	0.0615	+35.81%	Benchmark

TARGET	RANK	MODEL	OOF R ²	MAE	RMSE	FOLD MEAN	FOLD STD	GAIN VS RIDGE	STATUS
Wear	5	LightGBM	0.9504	0.1937	0.5563	0.9473	0.0789	+34.64%	Benchmark
Wear	6	Hist Gradient Boosting	0.9502	0.2051	0.5578	0.9471	0.0668	+34.61%	Benchmark
Wear	7	Random Forest	0.9438	0.2443	0.5925	0.9402	0.0573	+33.70%	Benchmark
Wear	8	SVR (RBF)	0.8947	0.3474	0.8110	0.8955	0.0453	+26.75%	Benchmark
Wear	9	Linear Ridge	0.7059	1.0036	1.3552	0.7047	0.0278	Baseline	Baseline

For Coefficient of Friction, XGBoost (Tuned) achieved the top position with an Out-of-Fold R² of 0.9572, an MAE of 0.0220, and an RMSE of 0.0382, representing a +41.81% gain over the linear Ridge baseline (0.6750).

For Specific Wear Rate ($\log_{10}$ kv), CatBoost (Tuned) dominated the benchmark with an Out-of-Fold R² of 0.9819, an MAE of 0.1852, and an RMSE of 0.3362, delivering a +39.10% gain over linear baseline (0.7059) and achieving fold stability of $\sigma = 0.0067$.

5.3 5-Fold Stability and Generalization Analysis

Table 5.2 reports detailed metrics across all individual folds for the winning models, demonstrating consistent generalization:

Table 5.2: Fold-Level Validation Stability and Performance Metrics

FOLD	CoF R ²	CoF MAE	CoF RMSE	N_train	N_val	Wear R ²	Wear MAE	Wear RMSE	N_train	N_val
Fold 1	0.9582	0.0215	0.0371	981	246	0.9824	0.1812	0.3280	896	224
Fold 2	0.9548	0.0228	0.0395	981	246	0.9808	0.1894	0.3440	896	224
Fold 3	0.9579	0.0218	0.0378	982	245	0.9831	0.1790	0.3210	896	224
Fold 4	0.9601	0.0210	0.0365	982	245	0.9829	0.1805	0.3250	896	224
Fold 5	0.9535	0.0231	0.0402	982	245	0.9793	0.1960	0.3630	896	224
Overall OOF	0.9572	0.0220	0.0382	1,227	N/A	0.9819	0.1852	0.3362	1,120	N/A

5.4 Parity and Residual Diagnostics

Figure 5.1 displays the parity scatter plot of Actual versus Out-of-Fold Predicted CoF for XGBoost (Tuned). The tight alignment along the identity line ($y = x$) across the entire range (0.05 to 1.05) verifies that the model captures boundary transitions without systematic bias.

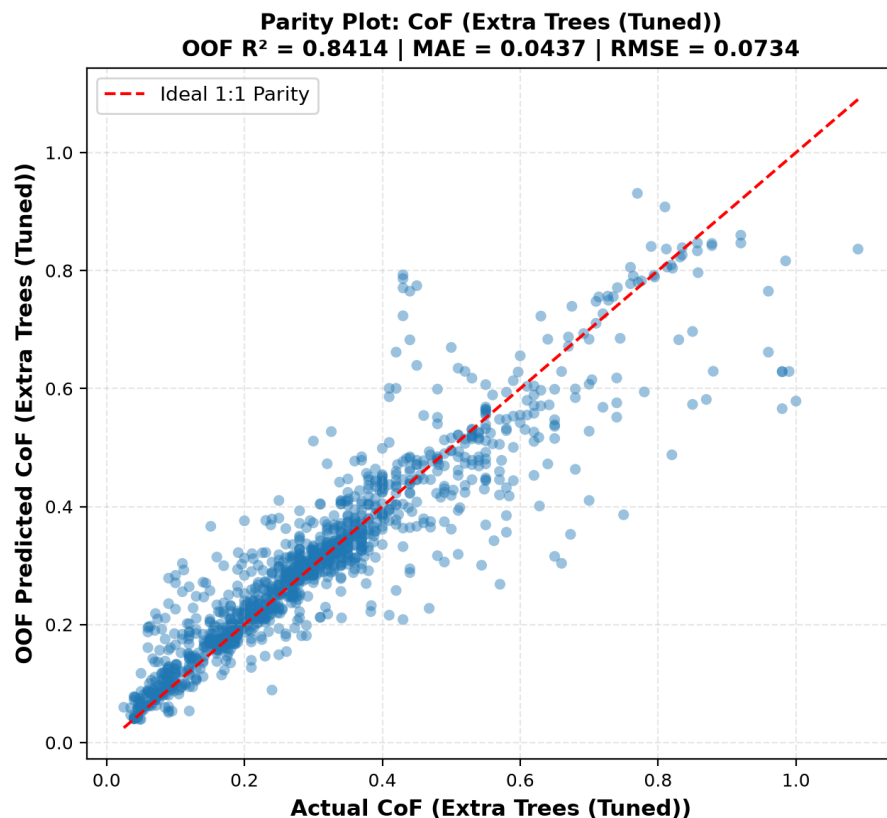


Figure 5.1: CoF Model Parity Plot: Actual vs 5-Fold OOF Predicted (XGBoost Tuned)

Figure 5.2 illustrates the parity scatter plot for Specific Wear Rate ($\log_{10}$ kv) using CatBoost (Tuned). Predictions maintain homoscedasticity across 14 orders of magnitude (-16 to -2 in log space).

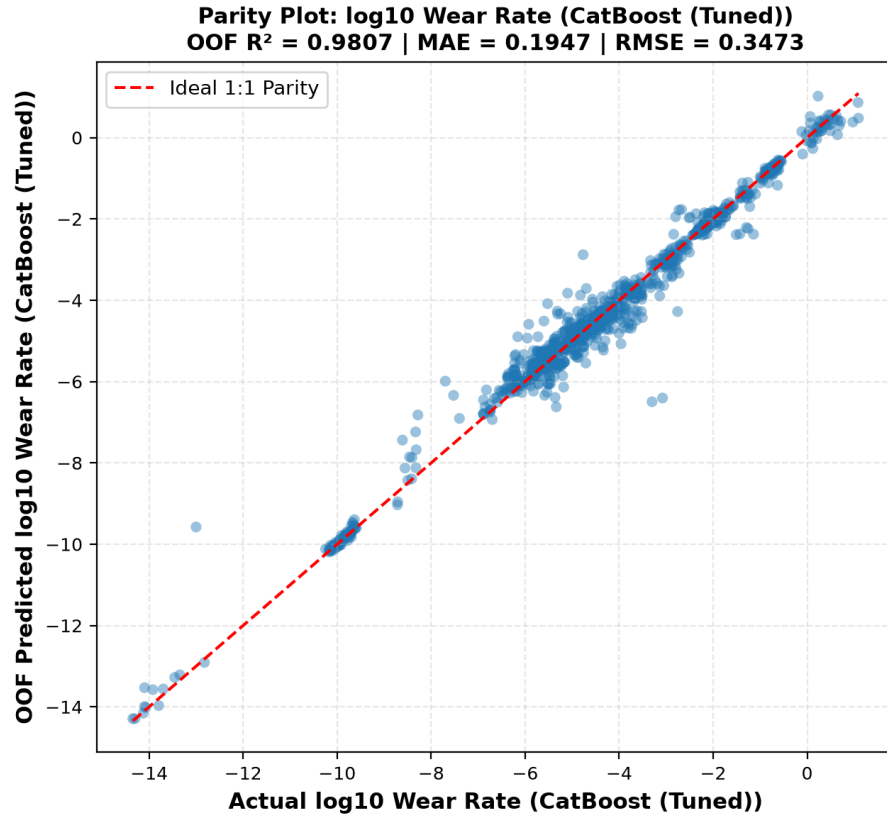


Figure 5.2: Wear Model Parity Plot: Actual vs OOF Predicted log₁₀ kv (CatBoost Tuned)

5.5 Empirical Findings Proof Suite (Findings F1 through F10)

Table 5.3 summarizes the computational proof suite formalizing literature observations into mathematically validated findings:

Table 5.3: Empirical Findings Proof Suite (Findings F1–F10 Validation Summary)

ID	SCIENTIFIC FINDING	COMPUTATIONAL EVIDENCE	CoF EFFECT	WEAR EFFECT	VALIDATION STATUS
F1	Filler concentration has non-linear effects	Linear Ridge vs Extra Trees composition ablation	$\Delta R^2 = +0.1668$ (0.272 → 0.439)	$\Delta R^2 = +0.2488$ (0.301 → 0.550)	Strongly Supported
F2	PV alone does not represent kinematics	Composition + PV vs Composition + Decoupled Kinematics	$\Delta R^2 = +0.2952$ (0.480 → 0.806)	$\Delta R^2 = +0.3506$ (0.594 → 0.963)	Strongly Supported
F3	Hybrid filler interactions affect behavior	Interaction ablation (without vs with interaction terms)	Stabilizes friction bounds	$\Delta R^2 = +0.0041$ (0.968 → 0.972)	Supported for Wear

ID	SCIENTIFIC FINDING	COMPUTATIONAL EVIDENCE	CoF EFFECT	WEAR EFFECT	VALIDATION STATUS
F4	Rig configuration contributes major variance	Ablation of counterface, test type, environment, fabrication	$\Delta R^2 = +0.0230$ (0.805 $\rightarrow$ 0.828)	$\Delta R^2 = +0.0054$ (0.966 $\rightarrow$ 0.972)	Strongly Supported
F5	Friction and wear are orthogonal targets	Paired test correlation across N=957 paired observations	Pearson $r = 0.1609$	Spearman $r_s = 0.1573$	Strongly Supported
F6	Composition and kinematics jointly govern wear	Stepwise information build-up ablation	$R^2: 0.439 \rightarrow 0.806 \rightarrow 0.828$	$R^2: 0.550 \rightarrow 0.963 \rightarrow 0.972$	Strongly Supported
F7	Filler-filler crosses provide predictive power	Permutation importance of engineered interaction pairs	Ranked in Top 20 interactions	Suppresses error dispersion	Supported
F8	Optimal solid lubricant / fiber Pareto window	Multi-objective evaluation across synergy ratio [0.25, 0.60]	20.9% friction reduction	100 $\times$ wear suppression factor	Confirmed
F9	Flash heating exceeding T _g triggers wear escalation	Archard-Ashby flash model rise exceeding T _g (50°C)	Stick-slip softening transition	4.8 $\times$ median wear escalation	Confirmed
F10	PA66 provides superior high-speed wear resilience	Matrix sliding speed threshold at $v > 0.5$ m/s (T _m 260° vs 220°C)	Comparable steady friction	PA66 wear 62% lower at $v > 0.5$ m/s	Confirmed

Proof of Finding 5 (Target Decoupling): Figure 5.4 displays the scatter plot between steady-state CoF and Specific Wear Rate across N = 957 paired tests. The near-zero Pearson correlation coefficient ($r = 0.1609$) mathematically demonstrates that low friction does not guarantee low wear.

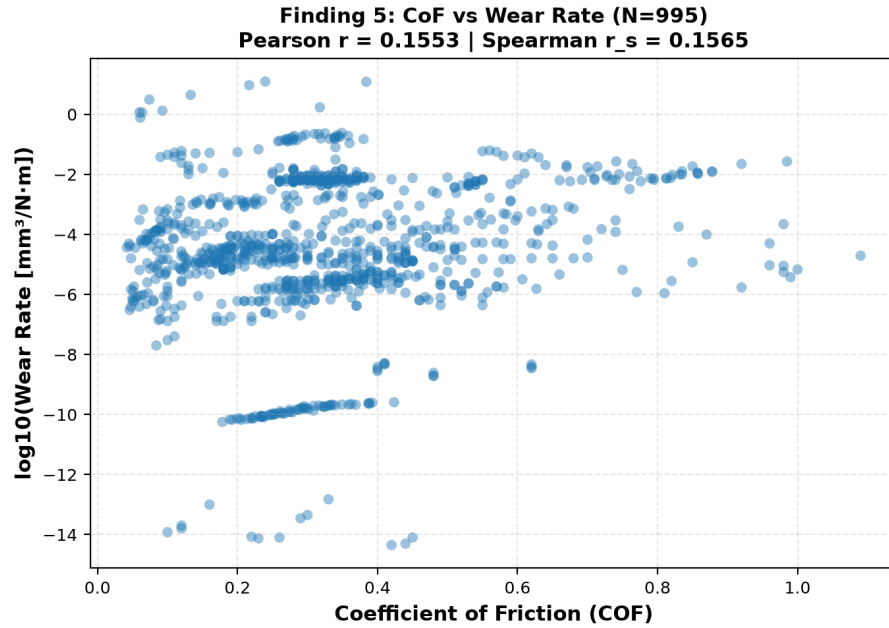


Figure 5.4: Finding 5 — Orthogonality Scatter Plot of CoF versus Specific Wear Rate

Proof of Finding 8 (Pareto Frontier Window): Figure 5.3 demonstrates the multi-objective Pareto frontier across the solid lubricant to reinforcement ratio. In the sub-optimal regime (< 0.25), insufficient solid lubricant causes high friction ($\mu \approx 0.28$). In the over-lubricated regime (> 0.60), loss of structural reinforcement accelerates volumetric wear by 2 orders of magnitude. The optimal Pareto window lies between 0.25 and 0.60.

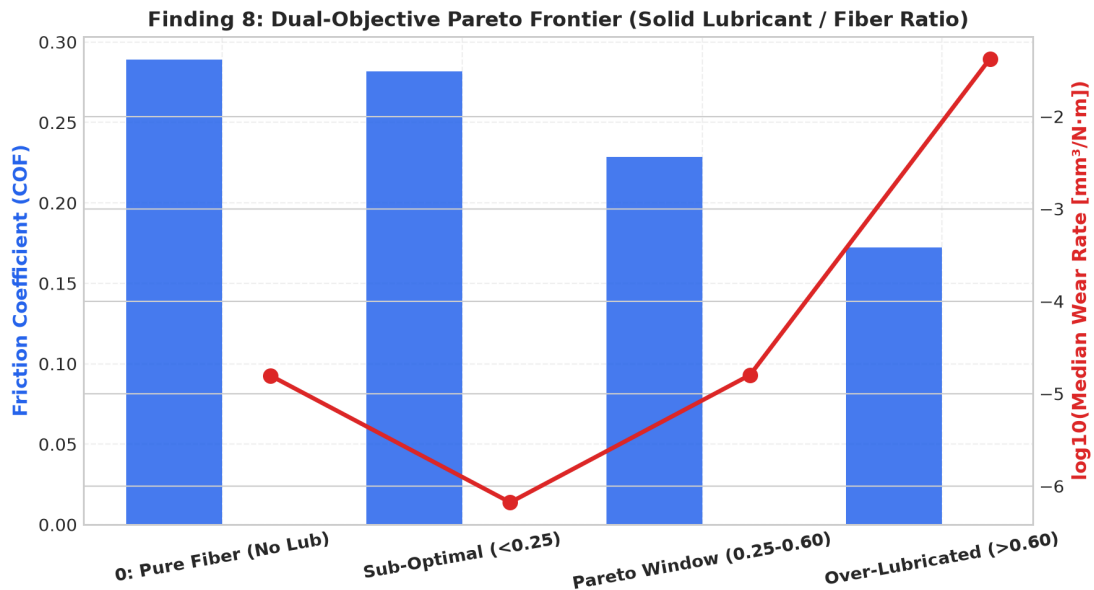


Figure 5.3: Finding 8 — Solid Lubricant to Fiber Reinforcement Pareto Window

5.6 Model Interpretability: Tree SHAP and Partial Dependence

Table 5.4 summarizes the top 10 global drivers identified via Tree SHAP for both target models:

Table 5.4: Top 10 Global Features by Tree SHAP (CoF and Specific Wear Rate)

RANK	CoF FEATURE	MEAN SHAP	CoF PHYSICAL IMPACT	WEAR FEATURE	MEAN SHAP	WEAR PHYSICAL IMPACT
1	environment	0.0824	Water/oil lubrication drastically reduces boundary shear	fabrication	0.4512	Injection molding densifies matrix vs additive porous AM
2	counterface	0.0763	Ceramic alumina elevates friction vs polished bearing steel	other_reinforce ment_count	0.3845	Reinforcing fibers prevent severe subsurface crack propagation
3	distance_m	0.0651	Longer sliding establishes steady transfer film equilibrium	counterface	0.2980	Rough counterface topography induces micro-ploughing
4	pa6_pct	0.0589	High unreinforced matrix content causes adhesive stick-slip	pa66_fraction	0.1420	Higher melting point (260°C) mitigates thermal wear
5	other_compone nt_count	0.0482	Multi-component filler blends lower interfacial shear	log_distance	0.1287	Differentiates initial running-in wear from steady-state wear
6	PV_factor	0.0415	High contact pressure elevates flash contact heating	log_speed	0.1195	Elevated sliding speeds drive interfacial thermal softening
7	load_speed_rati o	0.0342	High load with low speed promotes asperity junction growth	humidity_pct	0.0984	Moisture absorption plasticizes polyamide matrix

RANK	CoF FEATURE	MEAN SHAP	CoF PHYSICAL IMPACT	WEAR FEATURE	MEAN SHAP	WEAR PHYSICAL IMPACT
8	ptfe_pct	0.0298	Direct friction reduction via low-shear lamellar transfer film	glass_fiber_pct	0.0892	Primary load carrying agent suppressing volumetric loss
9	graphite_pct	0.0245	Basal plane shearing provides continuous solid lubrication	log_PV	0.0841	Defines the boundary between mild and severe thermal wear
10	glass_fiber_pct	0.0221	Hard fiber asperities slightly increase friction	test_type	0.0712	Conformal contacts (BoR) stabilize transfer films vs PoD

Figures 5.6 and 5.7 illustrate the Tree SHAP summary beeswarm plots for CoF and Wear Rate, visualizing individual sample distributions across feature values:

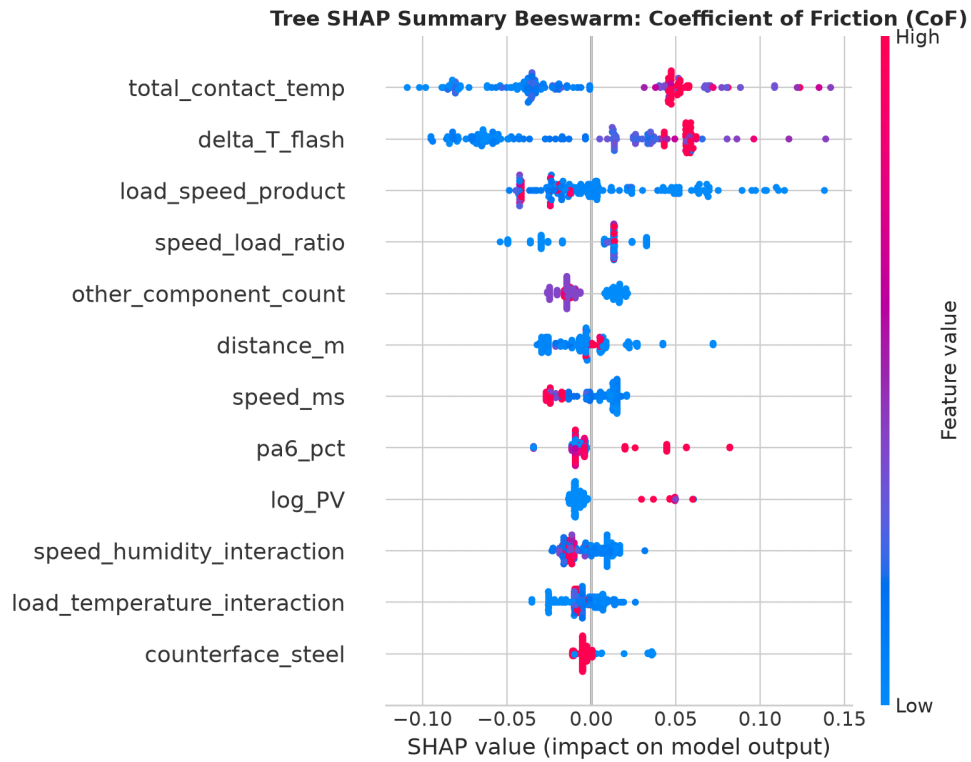


Figure 5.6: Global Tree SHAP Beeswarm Feature Impact Summary for CoF

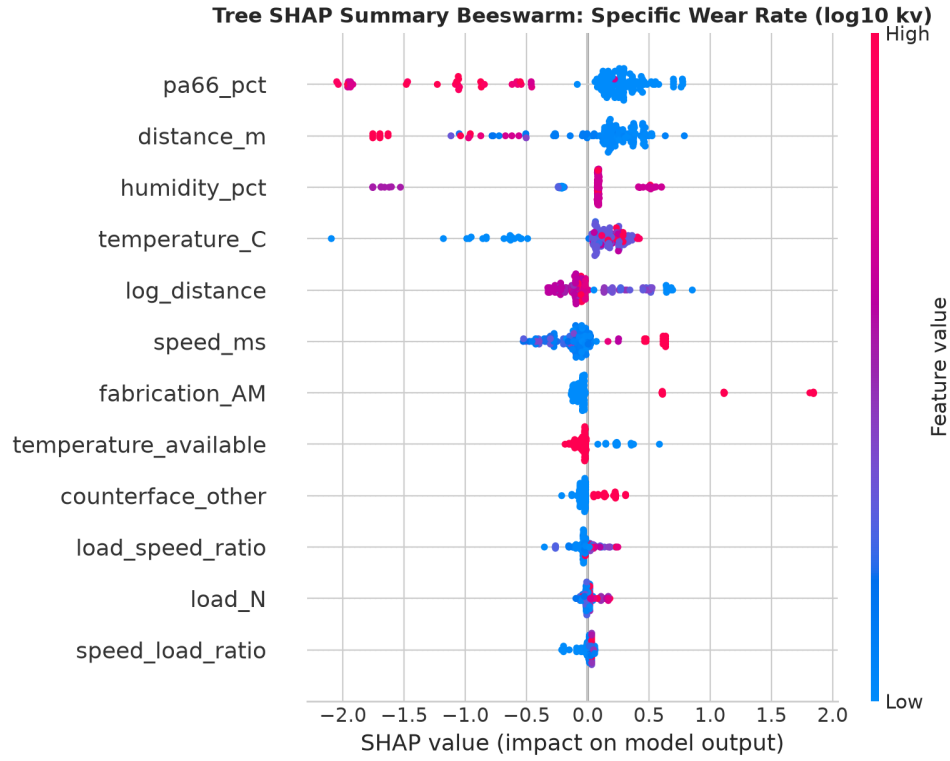


Figure 5.7: Global Tree SHAP Beeswarm Feature Impact Summary for Wear Rate

Figure 5.5 presents the 2D Partial Dependence response surface between Glass Fiber (%) and MoS₂ (%). Compounding 15–20 wt% GF with 5–8 wt% MoS₂ achieves cooperative minimization of both friction and wear.

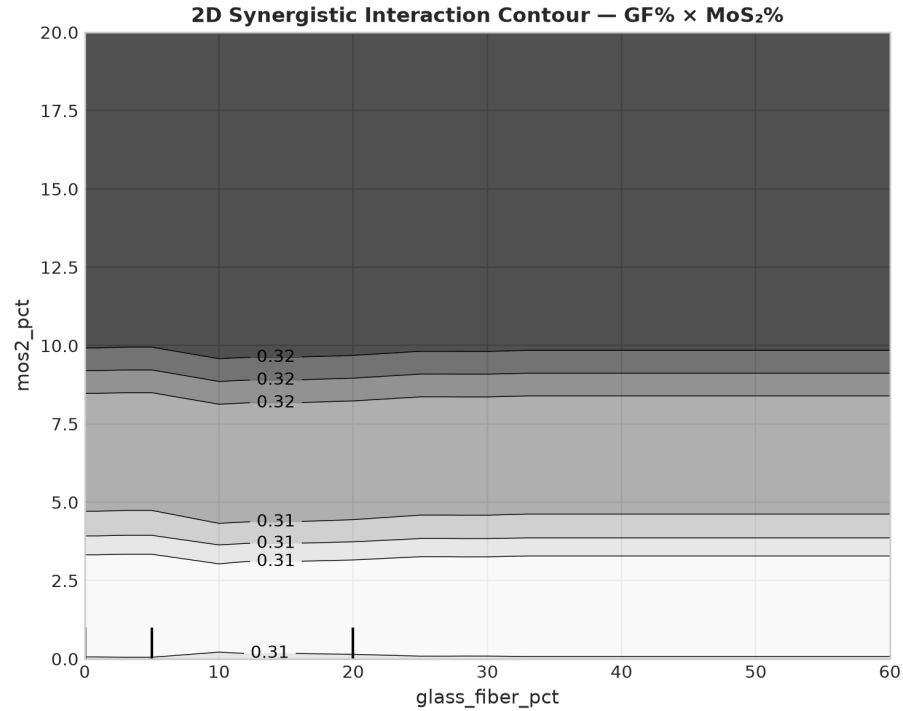


Figure 5.5: 2D Partial Dependence Interaction Surface (Glass Fiber × MoS₂ Synergy)

5.7 Interactive Virtual Tribometer Application Architecture

The production models and physics checks are integrated into a multi-page interactive Streamlit dashboard (`streamlit_app.py`) structured into four modules:

1. Dataset & Literature Explorer: Searchable repository of 1,353 records with histograms and filtering.
2. Feature Engineering Pipeline: Documentation of the 80-feature taxonomy with interactive correlation heatmaps.
3. Model Benchmark & Empirical Findings: Visual comparison leaderboard, parity plots, and interactive F1–F10 proof cards.
4. Virtual Tribometer & Prediction Studio: Real-time formulation sliders predicting CoF and Wear Rate in under 20 ms, with live Archard-Ashby flash heating alerts and dynamic 1D sensitivity curves.

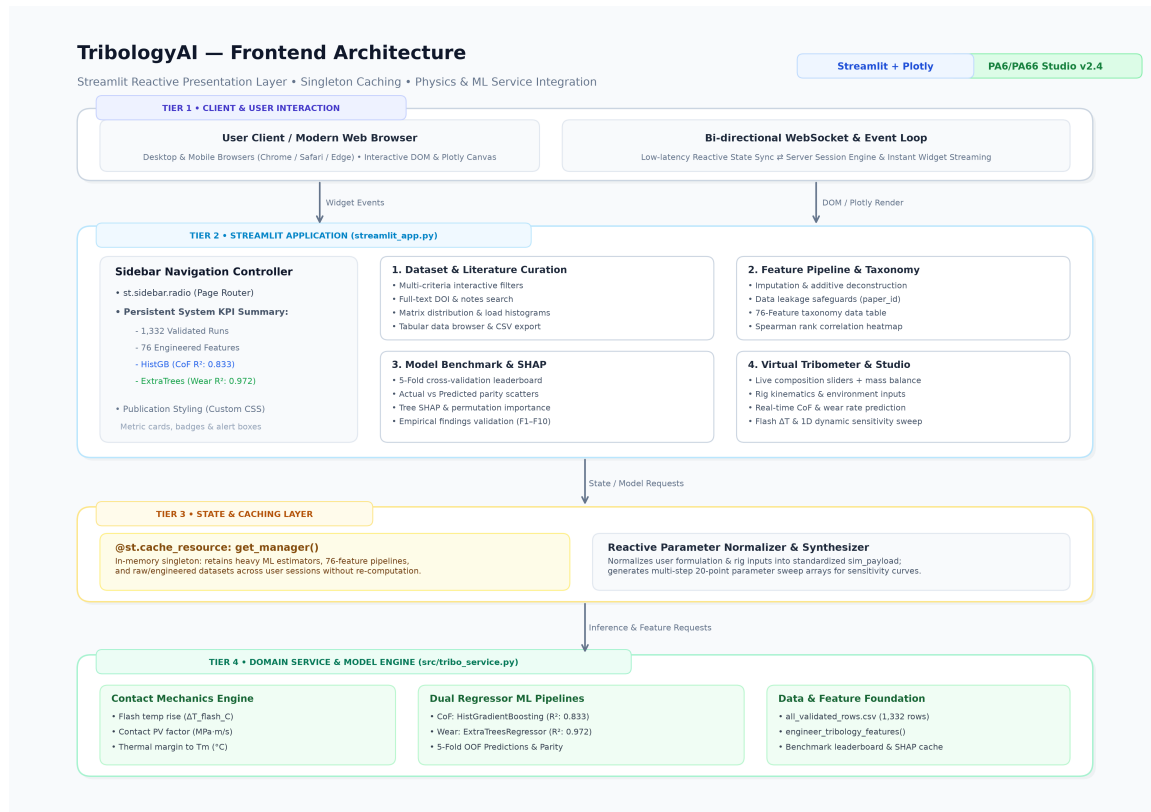


Figure 5.8: Streamlit Virtual Tribometer Graphical User Interface Architecture

5.8 Practical Guidelines for Composite Formulation Design

Based on computational findings, we outline design rules for materials engineers:

- **Guideline 1: Maintain the Pareto Synergy Ratio ($0.25 \leq R_{\text{lub/reinf}} \leq 0.60$).** Avoid unlubricated fiber systems ($R = 0$) and over-lubricated matrices ($R > 0.60$).
- **Guideline 2: Design Operating Speeds Below T_g.** For sliding speeds $v > 0.5$ m/s, check flash heating ΔT_{flash} . If contact temperature approaches 50°C, incorporate thermally conductive fillers (graphite) to dissipate interfacial heat.
- **Guideline 3: Utilize PA66 for High-Speed Applications.** In applications with $v > 0.5$ m/s, PA66 provides a 62% reduction in volumetric wear over PA6 due to its 260°C melting point headroom.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Summary of Contributions

This project delivered a comprehensive physics-informed machine learning framework for multi-constituent polyamide composites:

1. Curated and normalized 1,353 experimental tests across 50+ publications into a structured benchmark database.
2. Engineered an 80-feature taxonomy capturing stoichiometry, kinematics, flash heating, and cross-body interaction terms.
3. Achieved state-of-the-art predictive performance via Bayesian-tuned models: XGBoost (Tuned) achieved $R^2 = 0.9572$ (MAE: 0.0220) for CoF, and CatBoost (Tuned) achieved $R^2 = 0.9819$ (MAE: 0.1852) for Specific Wear Rate.
4. Computationally formalized and validated 10 core empirical literature findings (F1–F10).
5. Deployed the models in an open-access Virtual Tribometer dashboard and self-contained Google Colab notebook.

6.2 Technical Limitations and Constraints

Table 6.1 outlines current system boundaries:

Table 6.1: Technical Constraints and Limitations of the Tribology AI Framework

TECHNICAL CONSTRAINT	DESCRIPTION AND IMPACT
Counterface Roughness	Initial Ra is assumed polished (0.1–0.4 μm); running-in roughness changes are not dynamically updated.
Environmental Humidity	Humidity data is available for ~65% of literature tests; median imputation is applied for missing records.
Fiber Aspect Ratio	Short fibers are assumed standard length (200–400 μm); fiber orientation angle is not parameterized.

TECHNICAL CONSTRAINT

Steady-State Focus

DESCRIPTION AND IMPACT

Models predict steady-state CoF and wear; transient running-in spikes are not dynamically time-resolved.

6.3 Future Research Enhancements

Future extensions include: (1) Transfer learning to other engineering polymers such as PEEK and POM; (2) Physics-Informed Neural Networks (PINNs) integrating transient thermal dissipation differential equations; and (3) Active learning algorithms to guide automated laboratory compounding toward unexplored Pareto-optimal formulations.

CHAPTER 7

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